LIBBY — TARGET VALIDATION PATHS

Target validation paths — tnbc-brca1-oligomet-liver-r7p3

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Six gating workups load the 1L conversation in parallel: germline BRCA1 / hereditary breast-ovarian panel (gates olaparib via NCT02000622 and talazoparib via NCT01945775 on FDA label; the somatic-only contingency routes to NCT03990896), PD-L1 22C3 CPS on the breast primary block (gates pembrolizumab + chemo via KEYNOTE-355 / NCT02819518, sacituzumab govitecan + pembrolizumab via ASCENT-04 / NCT05382299, and datopotamab deruxtecan ± durvalumab via TROPION-Breast05 / NCT06103864), TMB on F1CDx (gates tumor-agnostic pembrolizumab under KEYNOTE-158 / NCT02628067 for later lines), PIK3CA hotspot resolution (gates alpelisib, inavolisib, and capivasertib eligibility, even though the PI3K/AKT-axis evidence in TNBC reads negative or terminated), PTEN IHC (gates capivasertib at the protein-level call), and HER2 IHC 0 vs 1+ / 2+ FISH-negative (gates trastuzumab deruxtecan for later lines if HER2-low reflexes positive). None of these involves a fresh biopsy. They run on archival FFPE plus a peripheral blood draw, in parallel, with turnaround between five business days and three weeks.

BRCA1

Essential: germline BRCA1/2 sequencing plus del/dup on a hereditary breast-ovarian panel that also covers PALB2, ATM, CHEK2, BARD1, RAD51C/D, BRIP1, TP53, PTEN, CDH1, and STK11. The tumor NGS call does not say whether the BRCA1 variant is germline or somatic, and the answer changes a 42-year-old's life: germline carriers face bilateral mastectomy and risk-reducing salpingo-oophorectomy decisions and trigger cascade testing of first-degree relatives, while somatic-only BRCA1 still supports PARP-inhibitor and platinum sensitivity but does not drive surgical or family-screening choices. NCCN BINV and genetic-familial breast guidelines mandate this for any patient diagnosed under 50 or with TNBC. Pair the order with a cancer-genetics counseling visit so cascade testing can be triggered the same week the variant is classified. Turnaround is two to three weeks.

High priority: an HRD genomic-scar score (Myriad MyChoice CDx GIS, or the LOH-based signal on FoundationOne CDx) as orthogonal proof that the BRCA1 variant is producing a functional HR-deficient phenotype rather than sitting at a permissive VAF. Does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC, but it strengthens the platinum / PARP rationale and gives any future HR-targeted trial discussion a baseline scar reference.

High priority: a baseline ctDNA draw before any PARP-inhibitor or platinum exposure (Guardant360, FoundationOne Liquid CDx, or Natera Signatera tumor-informed). BRCA1/2 reversion mutations are detectable in roughly 60% of patients who progress on PARP inhibitors and are the dominant acquired-resistance mechanism in BRCA-mutated breast cancer. Without a clean baseline, the reversion call at later progression becomes ambiguous and the team cannot cleanly distinguish on-pathway resistance from off-pathway escape.

PD-L1 / TILs

Essential: PD-L1 IHC 22C3 pharmDx on the breast primary FFPE block (CPS scoring for pembrolizumab; SP142 IC for atezolizumab if needed). The pembrolizumab + chemotherapy label for metastatic TNBC requires CPS ≥10 by the 22C3 companion diagnostic, and PD-L1 expression in TNBC is discordant between primary breast and liver metastasis in roughly 20-30% of paired samples. Run on the breast primary; if only liver M1 is available, run on both and document the discordance rate. Turnaround is five to seven business days.

Medium priority: ITWG stromal-TIL re-scoring on H and E by a dedicated breast pathologist using the Salgado 2014 methodology on a 0-100% scale. The 3+ stromal-TIL label sits cleanly with ITWG only if the scoring was done as a continuous percentage on a single tumor-bearing section; ad-hoc 1+/2+/3+ grading inflates inter-observer variability. The re-score does not gate ICI eligibility (TIL-rich status is informative, not labeled) but it makes the TIL signal defensible at tumor board.

TMB

Essential: TMB measurement on an FDA-cleared comprehensive genomic panel sized ≥0.8 Mb, with the panel and threshold documented in the report. The tumor-agnostic pembrolizumab approval under KEYNOTE-158 is anchored on FoundationOne CDx as the companion diagnostic at a ≥10 mut/Mb cutoff, and TMB values are not interchangeable across panels. Smaller or differently-bioinformatic panels can over- or under-call TMB by 3-5 mut/Mb at the clinically relevant threshold. A 14 mut/Mb value from a non-FDA-cleared assay near the 10 mut/Mb cutoff is exactly the situation where the call should be locked to a panel the FDA label recognizes. If the original 14 mut/Mb already came from F1CDx, this row collapses to a chart-review check. Otherwise, re-run on F1CDx before using the value to gate pembrolizumab.

Medium priority: baseline B2M and HLA class I IHC (B2M EP2978Y, HLA-ABC EMR8-5) plus a re-read of B2M / HLA-A/B/C variant calls on the existing tumor NGS panel. B2M loss and HLA class I antigen-presentation defects are well-characterized primary-resistance mechanisms for ICI in TMB-high and MSI-H tumors. Cheap to add at the same time as the PD-L1 stain; refines the ICI rationale without gating it.

PIK3CA / PTEN

Essential: PIK3CA variant resolution to the specific codon (therascreen PIK3CA RGQ PCR Kit or FoundationOne CDx) covering C420R, E542K, E545A/D/G/K, Q546E/R, H1047L/R/Y. The FDA companion diagnostics for alpelisib and capivasertib only call a tumor PIK3CA-altered for this 11-variant hotspot set. A non-hotspot helical or kinase-domain variant does not get a patient onto an alpelisib-labeled regimen. If the existing NGS already names the codon, this collapses to a chart-review check; if it does not, a hotspot-aware confirmation is the rate-limiting step before any PI3K-axis discussion. Even though the TNBC PI3K/AKT-axis evidence reads negative (CAPItello-290 was negative overall and in the AKT-pathway-altered stratum; EPIK-B3 was terminated for slow recruitment), the gate clarifies pathway biology rather than unlocking an active recommendation here.

Essential: PTEN IHC (clone 138G6 or SP218) with a positive internal control, scored as deficient if less than 10% of tumor cells stain at any intensity. Copy-number loss on NGS does not always translate to PTEN protein loss: PTEN is frequently inactivated by promoter methylation, frameshift, or LOH that panel-level copy-number resolution can miss. The IHC re-stain is the orthogonal protein-level confirmation that pathologists and tumor boards expect before AKT-inhibitor discussion. Turnaround is five to seven business days.

Basal-like subtype

High priority: PAM50 intrinsic-subtype assay (NanoString Prosigna or Agendia BluePrint) with concurrent Lehmann TNBCtype-4 (BL1, BL2, M, LAR) classification on RNA-seq data. Roughly 25% of TNBCs do not classify as PAM50 basal-like, and within basal-like the Lehmann TNBCtype-4 subdivision carries divergent neoadjuvant pCR rates: BL1 about 52%, BL2 about 0%, LAR about 10%, M about 23% under anthracycline-taxane backbones. The current basal-like call should be locked to the assay that made it. The Lehmann layer adds therapeutic resolution that the basal-vs-non-basal binary cannot: LAR enrichment shifts the conversation toward AR-directed agents, M or MSL shifts it away from KEYNOTE-style ICI + taxane, and BL1 reinforces the platinum / PARP backbone already supported by BRCA1.

TP53 / MYC / IRS2 (context)

Medium priority: a comprehensive NGS review for replication-stress and DDR co-alterations (ATM, ATR, CHK1/2, CDK12, CCNE1 amplification) and TP53 variant-functional classification. TP53 mutation is near-universal in basal-like TNBC and not directly drug-gating, but the co-alteration profile anchors the WEE1-inhibitor and ATR-inhibitor trial rationale. Shared block with the PIK3CA, PTEN, and TMB orders.

Medium priority: MYC FISH (8q24 break-apart / amplification probe set) or MYC IHC (clone Y69) on archival FFPE. MYC amplification by NGS copy-number is reasonably specific but the magnitude (low-level gain vs focal high-level amplification) varies by panel and tumor purity, and BET / CDK7 / CDK9 trial enrollment favors high-level amplification rather than borderline gain. Does not gate any approved therapy; refines MYC-pathway trial eligibility.

Low priority: IRS2 RNA expression on tumor RNA-seq with reference to the TCGA TNBC distribution. Copy-number amplification only matters clinically if it translates to elevated IRS2 transcript and a functional IGF-1R / IRS signaling readout. Deferrable until an IGF-1R-axis protocol is on the table. Bundle on the same RNA-seq order as PAM50 and Lehmann to share cost.

Where to order these assays

Assay	Provider	Decision gated	Contact
Germline BRCA1/2 + hereditary breast-ovarian panel	**Labcorp Genetics (formerly Invitae) (preferred) (Invitae Hereditary Breast and Gyn Cancers Panel)**	**Olaparib via NCT02000622 and talazoparib via NCT01945775 FDA-label eligibility; risk-reducing surgery (bilateral mastectomy, RRSO); cascade testing of first-degree relatives.**	**test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037**
Germline BRCA1/2 + hereditary breast-ovarian panel	Myriad Genetics (MyRisk Hereditary Cancer Panel)	Olaparib via NCT02000622 and talazoparib via NCT01945775 FDA-label eligibility; risk-reducing surgery (bilateral mastectomy, RRSO); cascade testing of first-degree relatives.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline BRCA1/2 + hereditary breast-ovarian panel	GeneDx (GeneDx Comprehensive Common Cancer Panel)	Olaparib via NCT02000622 and talazoparib via NCT01945775 FDA-label eligibility; risk-reducing surgery (bilateral mastectomy, RRSO); cascade testing of first-degree relatives.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline BRCA1/2 + hereditary breast-ovarian panel	Ambry Genetics (CancerNext)	Olaparib via NCT02000622 and talazoparib via NCT01945775 FDA-label eligibility; risk-reducing surgery (bilateral mastectomy, RRSO); cascade testing of first-degree relatives.	test info · 15 Argonaut, Aliso Viejo, CA 92656 · 1-866-262-7943

PD-L1 IHC 22C3 pharmDx on breast primary FFPE	**Labcorp Oncology (preferred) (PD-L1 IHC 22C3 pharmDx (Dako/Agilent))**	**Pembrolizumab + chemotherapy via KEYNOTE-355 (NCT02819518) at CPS ≥10; sacituzumab govitecan + pembrolizumab via ASCENT-04 (NCT05382299); datopotamab deruxtecan ± durvalumab via TROPION-Breast05 (NCT06103864).**	**test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363**
PD-L1 IHC 22C3 pharmDx on breast primary FFPE	Quest Diagnostics (PD-L1 22C3 pharmDx)	Pembrolizumab + chemotherapy via KEYNOTE-355 (NCT02819518) at CPS ≥10; sacituzumab govitecan + pembrolizumab via ASCENT-04 (NCT05382299); datopotamab deruxtecan ± durvalumab via TROPION-Breast05 (NCT06103864).	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
PD-L1 IHC 22C3 pharmDx on breast primary FFPE	NeoGenomics Laboratories (PD-L1 22C3 pharmDx)	Pembrolizumab + chemotherapy via KEYNOTE-355 (NCT02819518) at CPS ≥10; sacituzumab govitecan + pembrolizumab via ASCENT-04 (NCT05382299); datopotamab deruxtecan ± durvalumab via TROPION-Breast05 (NCT06103864).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
PD-L1 IHC 22C3 pharmDx on breast primary FFPE	Mayo Clinic Laboratories (PD-L1 22C3 pharmDx)	Pembrolizumab + chemotherapy via KEYNOTE-355 (NCT02819518) at CPS ≥10; sacituzumab govitecan + pembrolizumab via ASCENT-04 (NCT05382299); datopotamab deruxtecan ± durvalumab via TROPION-Breast05 (NCT06103864).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
TMB on FDA-cleared comprehensive genomic panel ≥0.8 Mb (FoundationOne CDx, MSK-IMPACT, or Tempus xT CDx)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Tumor-agnostic pembrolizumab under KEYNOTE-158 (NCT02628067) at ≥10 mut/Mb; TMB-restricted trial enrollment.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
TMB on FDA-cleared comprehensive genomic panel ≥0.8 Mb	Tempus Labs (Tempus xT CDx)	Tumor-agnostic pembrolizumab under KEYNOTE-158 (NCT02628067) at ≥10 mut/Mb; TMB-restricted trial enrollment.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
TMB on FDA-cleared comprehensive genomic panel ≥0.8 Mb	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Tumor-agnostic pembrolizumab under KEYNOTE-158 (NCT02628067) at ≥10 mut/Mb; TMB-restricted trial enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000

TMB on FDA-cleared comprehensive genomic panel ≥0.8 Mb	Caris Life Sciences (MI Cancer Seek)	Tumor-agnostic pembrolizumab under KEYNOTE-158 (NCT02628067) at ≥10 mut/Mb; TMB-restricted trial enrollment.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
PIK3CA variant resolution to specific codon (11-variant hotspot set)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Alpelisib, inavolisib, and capivasertib eligibility; trial enrollment for PI3K/AKT-axis studies requiring a hotspot-defined PIK3CA call.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
PIK3CA variant resolution to specific codon	Labcorp Oncology (therascreen PIK3CA RGQ PCR Kit (FDA-approved))	Alpelisib, inavolisib, and capivasertib eligibility; trial enrollment for PI3K/AKT-axis studies requiring a hotspot-defined PIK3CA call.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
PIK3CA variant resolution to specific codon	Tempus Labs (Tempus xT CDx)	Alpelisib, inavolisib, and capivasertib eligibility; trial enrollment for PI3K/AKT-axis studies requiring a hotspot-defined PIK3CA call.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
PIK3CA variant resolution to specific codon	Caris Life Sciences (MI Cancer Seek)	Alpelisib, inavolisib, and capivasertib eligibility; trial enrollment for PI3K/AKT-axis studies requiring a hotspot-defined PIK3CA call.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
PIK3CA variant resolution to specific codon	Guardant Health (Guardant360 CDx)	Alpelisib, inavolisib, and capivasertib eligibility; trial enrollment for PI3K/AKT-axis studies requiring a hotspot-defined PIK3CA call.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
PTEN IHC (clone 138G6 or SP218) with internal control, deficient if less than 10% tumor staining	**Mayo Clinic Laboratories (preferred) (PTEN Immunostain)**	**Capivasertib + fulvestrant (HR+/HER2-) decision; trial enrollment for AKT-inhibitor protocols in TNBC requiring PTEN-deficient status by IHC.**	**test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710**
PTEN IHC	NeoGenomics Laboratories (PTEN IHC)	Capivasertib + fulvestrant (HR+/HER2-) decision; trial enrollment for AKT-inhibitor protocols in TNBC requiring PTEN-deficient status by IHC.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
PTEN IHC	Memorial Sloan Kettering Pathology Consultation Service	Capivasertib + fulvestrant (HR+/HER2-) decision; trial enrollment for AKT-inhibitor protocols in TNBC requiring PTEN-deficient status by IHC.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
PTEN IHC	Labcorp / Esoterix (PTEN IHC)	Capivasertib + fulvestrant (HR+/HER2-) decision; trial enrollment for AKT-inhibitor protocols in TNBC requiring PTEN-deficient status by IHC.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

HRD genomic-scar score (Myriad MyChoice CDx GIS or LOH-based on FoundationOne CDx)	**Myriad Genetics (preferred) (MyChoice CDx (HRD with Genomic Instability Score))**	**Orthogonal HR-deficient phenotype call; strengthens platinum / PARP rationale; does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC.**	**test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423**
HRD genomic-scar score	Foundation Medicine (FoundationOne CDx (LOH-based HRD signal))	Orthogonal HR-deficient phenotype call; strengthens platinum / PARP rationale; does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
HRD genomic-scar score	Caris Life Sciences (MI Profile HRD)	Orthogonal HR-deficient phenotype call; strengthens platinum / PARP rationale; does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
HRD genomic-scar score	Tempus Labs (Tempus HRD)	Orthogonal HR-deficient phenotype call; strengthens platinum / PARP rationale; does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
PAM50 intrinsic-subtype assay plus Lehmann TNBCtype-4 classification	**Veracyte (preferred) (Prosigna PAM50)**	**TNBC-subtype-directed therapy bias (AR-axis vs platinum / PARP vs ICI + taxane); trial-enrollment refinement for subtype-restricted protocols.**	**test info · 6000 Shoreline Court, Suite 300, South San Francisco, CA 94080 · 1-650-243-6300**
PAM50 intrinsic-subtype assay	Agendia (BluePrint 80-gene molecular subtyping)	TNBC-subtype-directed therapy bias (AR-axis vs platinum / PARP vs ICI + taxane); trial-enrollment refinement for subtype-restricted protocols.	test info · 22 Morgan, Suite 100, Irvine, CA 92618 · 1-888-321-2732
PAM50 intrinsic-subtype assay	Tempus Labs (Tempus xR (RNA-seq))	TNBC-subtype-directed therapy bias (AR-axis vs platinum / PARP vs ICI + taxane); trial-enrollment refinement for subtype-restricted protocols.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
PAM50 intrinsic-subtype assay	Caris Life Sciences (MI Cancer Seek + WTS)	TNBC-subtype-directed therapy bias (AR-axis vs platinum / PARP vs ICI + taxane); trial-enrollment refinement for subtype-restricted protocols.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Baseline ctDNA for BRCA1 reversion surveillance	**Guardant Health (preferred) (Guardant360 CDx)**	**Acquired PARP-inhibitor and platinum resistance surveillance; informs whether to push through resistance or switch to a non-HR-axis therapy.**	**test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887**

Baseline ctDNA for BRCA1 reversion surveillance	Foundation Medicine (FoundationOne Liquid CDx)	Acquired PARP-inhibitor and platinum resistance surveillance; informs whether to push through resistance or switch to a non-HR-axis therapy.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Baseline ctDNA for BRCA1 reversion surveillance	Natera (Signatera (tumor-informed ctDNA))	Acquired PARP-inhibitor and platinum resistance surveillance; informs whether to push through resistance or switch to a non-HR-axis therapy.	test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-249-9090
Baseline ctDNA for BRCA1 reversion surveillance	Tempus Labs (Tempus xF)	Acquired PARP-inhibitor and platinum resistance surveillance; informs whether to push through resistance or switch to a non-HR-axis therapy.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
ITWG stromal-TIL re-scoring (Salgado 2014 methodology, 0-100% scale)	**Memorial Sloan Kettering Pathology Consultation Service (preferred)**	**ICI-combination rationale strength (KEYNOTE-522, IMpassion-style); reinforces but does not gate enrollment.**	**test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511**
ITWG stromal-TIL re-scoring	Dana-Farber / Brigham and Women's Pathology	ICI-combination rationale strength (KEYNOTE-522, IMpassion-style); reinforces but does not gate enrollment.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510
ITWG stromal-TIL re-scoring	MD Anderson Department of Pathology	ICI-combination rationale strength (KEYNOTE-522, IMpassion-style); reinforces but does not gate enrollment.	test info · 1515 Holcombe Boulevard, Houston, TX 77030 · 1-877-632-6789
ITWG stromal-TIL re-scoring	Mayo Clinic Anatomic Pathology Consultation	ICI-combination rationale strength (KEYNOTE-522, IMpassion-style); reinforces but does not gate enrollment.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
MYC FISH (8q24 break-apart / amplification probe set) or MYC IHC (clone Y69)	**NeoGenomics Laboratories (preferred) (MYC FISH)**	**Refines MYC-amplification call strength for BET, CDK7, and CDK9 trial enrollment; does not gate approved therapy.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
MYC FISH or IHC	Mayo Clinic Laboratories (MYCBA / MYC Break-Apart FISH)	Refines MYC-amplification call strength for BET, CDK7, and CDK9 trial enrollment; does not gate approved therapy.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
MYC FISH or IHC	Labcorp / Esoterix (MYC FISH)	Refines MYC-amplification call strength for BET, CDK7, and CDK9 trial enrollment; does not gate approved therapy.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
MYC FISH or IHC	Quest Diagnostics (MYC FISH)	Refines MYC-amplification call strength for BET, CDK7, and CDK9 trial enrollment; does not gate approved therapy.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

B2M and HLA class I IHC plus variant-call review on tumor NGS	**NeoGenomics Laboratories (preferred) (B2M IHC / HLA Class I IHC)**	**Primary-resistance risk for pembrolizumab and other PD-1/PD-L1 agents; refines the ICI rationale without gating it.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
B2M and HLA class I IHC	Mayo Clinic Laboratories (B2M / HLA Class I IHC)	Primary-resistance risk for pembrolizumab and other PD-1/PD-L1 agents; refines the ICI rationale without gating it.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
B2M and HLA class I IHC	Memorial Sloan Kettering Pathology Consultation Service	Primary-resistance risk for pembrolizumab and other PD-1/PD-L1 agents; refines the ICI rationale without gating it.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
Comprehensive NGS review for replication-stress / DDR co-alterations and TP53 variant classification	**Foundation Medicine (preferred) (FoundationOne CDx)**	**WEE1 / ATR-inhibitor trial enrollment rationale; not gating for any approved therapy.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Comprehensive NGS review for DDR co-alterations	Tempus Labs (Tempus xT CDx)	WEE1 / ATR-inhibitor trial enrollment rationale; not gating for any approved therapy.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Comprehensive NGS review for DDR co-alterations	Caris Life Sciences (MI Cancer Seek)	WEE1 / ATR-inhibitor trial enrollment rationale; not gating for any approved therapy.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
IRS2 RNA expression on tumor RNA-seq	**Tempus Labs (preferred) (Tempus xR (whole-transcriptome RNA-seq))**	**IGF-1R / IRS-axis trial enrollment; does not gate approved therapy.**	**test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137**
IRS2 RNA expression	Caris Life Sciences (MI Profile WTS)	IGF-1R / IRS-axis trial enrollment; does not gate approved therapy.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

PIK3CA variant resolution to specific codon (therascreen PIK3CA RGQ PCR Kit hotspots or FoundationOne CDx) covering C420R, E542K, E545A/D/G/K, Q546E/R, H1047L/R/Y	The profile says PIK3CA is mutated but does not specify the codon, and the FDA companion diagnostics for alpelisib (therascreen PIK3CA) and the F1CDx label for capivasertib (PIK3CA/AKT1/PTEN) only call a tumor PIK3CA-altered for the 11-variant hotspot set. A non-hotspot helical or kinase-domain variant does not get a patient onto an alpelisib-labeled regimen and may or may not be accepted by capivasertib protocols, so the resolution gates which PI3K/AKT-axis drug can be offered at all. If the existing NGS already names the codon, this row collapses to a documentation check; if it does not, a hotspot-aware confirmation is the rate-limiting step before any PI3K-axis discussion.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable

PTEN IHC (clone 138G6 or SP218) with positive internal control, scored as deficient if less than 10% tumor-cell staining at any intensity	Copy-number loss on NGS does not always translate to PTEN protein loss; PTEN is frequently inactivated by promoter methylation, frameshift, or LOH not picked up by panels with limited copy-number resolution. The FDA approval of capivasertib (CAPItello-291) and the F1CDx companion-diagnostic label define PTEN-altered on a DNA call, but the IHC re-stain with a less-than-10% cutoff is the orthogonal protein-level confirmation that pathologists and tumor boards expect before AKT-inhibitor discussion. Skipping it leaves the AKT-inhibitor rationale resting on a single copy-number signal without protein-level corroboration.	Mayo Clinic Laboratories (PTEN Immunostain (technical component)) · test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710	1 unstained slide from archival FFPE
TMB measurement on an FDA-cleared comprehensive genomic panel sized ≥0.8 Mb (FoundationOne CDx, MSK-IMPACT, or Tempus xT CDx), with the panel and threshold documented in the report	The tumor-agnostic pembrolizumab approval (KEYNOTE-158) is anchored on FoundationOne CDx as the companion diagnostic at a ≥10 mut/Mb cutoff, and TMB values are not interchangeable across panels: smaller or differently-bioinformatic panels can over- or under-call TMB by 3-5 mut/Mb at the clinically relevant threshold. A 14 mut/Mb value from a non-FDA-cleared assay near the 10 mut/Mb cutoff is exactly the situation where the call should be locked to a panel the FDA label recognizes before pembrolizumab is offered on a TMB-only basis.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable

PD-L1 IHC 22C3 pharmDx CPS on the breast primary block	In metastatic TNBC the pembrolizumab + chemotherapy label (KEYNOTE-355) requires PD-L1 CPS ≥10 by the 22C3 pharmDx companion diagnostic. PD-L1 expression in TNBC is discordant between primary breast and liver metastasis in roughly 20-30% of paired samples, so order on the breast primary; if only liver M1 is available, run on both and document the discordance rate. Skipping it means defaulting to PD-L1-agnostic backbones and missing the chance to anchor a CPS ≥10 case onto the pembrolizumab + chemo regimen.	Labcorp Oncology (PD-L1 IHC 22C3 pharmDx (Dako/Agilent)) · test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363	1 FFPE block or 4-6 unstained slides
Germline BRCA1/2 sequencing plus del/dup on a hereditary breast/ovarian cancer panel covering PALB2, ATM, CHEK2, BARD1, RAD51C/D, BRIP1, TP53, PTEN, CDH1, STK11	The tumor NGS call does not say whether the BRCA1 variant is germline or somatic, and the answer changes a 42-year-old patient's life: germline carriers face bilateral mastectomy and risk-reducing salpingo-oophorectomy decisions and trigger cascade testing of first-degree relatives, while purely somatic BRCA1 still supports PARP-inhibitor and platinum sensitivity but does not drive surgical or family-screening choices. NCCN BINV and genetic-familial breast guidelines mandate germline testing in any patient diagnosed under 50 or with TNBC regardless of age.	Labcorp Genetics (formerly Invitae) (Invitae Hereditary Breast and Gyn Cancers Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	5-10 mL whole blood (EDTA) or saliva kit

HRD genomic-scar score (Myriad MyChoice CDx GIS or equivalent on FoundationOne CDx / Caris MI Profile)	An HRD score (LOH + telomeric allelic imbalance + large-scale state transitions) gives orthogonal proof that the BRCA1 variant is producing a functional HR-deficient phenotype, not just sitting at a permissive VAF. This matters most when the BRCA1 call sits in a domain whose functional consequence is debatable, or when later resistance lines need a baseline scar score to interpret BRCA reversion in ctDNA. The score does not gate PARP-inhibitor eligibility in BRCA1-mutated TNBC (that runs off the BRCA call itself), but it strengthens the platinum / PARPi rationale.	Myriad Genetics (MyChoice CDx (HRD with Genomic Instability Score)) · test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423	1 FFPE block or 10-20 unstained slides
PAM50 intrinsic-subtype assay (NanoString Prosigna or Agendia BluePrint) with concurrent Lehmann TNBCtype-4 classification on RNA-seq	Roughly 25% of TNBCs do not classify as PAM50 basal-like, and within basal-like the Lehmann TNBCtype-4 subdivision carries divergent neoadjuvant pCR rates: BL1 about 52%, BL2 about 0%, LAR about 10%, M about 23% under anthracycline-taxane backbones. The current basal-like call should be locked to the assay that made it (PAM50 vs ad-hoc gene set), and the Lehmann layer adds therapeutic resolution that the basal-vs-non-basal binary cannot.	Veracyte (Prosigna PAM50) · test info · 6000 Shoreline Court, Suite 300, South San Francisco, CA 94080 · 1-650-243-6300	FFPE block or 10 unstained slides

Baseline ctDNA for BRCA1 reversion / secondary mutation surveillance	BRCA1/2 reversion mutations are detectable in ctDNA in roughly 60% of patients who progress on PARP inhibitors and are the single most common acquired-resistance mechanism in BRCA-mutated breast cancer. A baseline draw before any PARP-inhibitor or platinum exposure gives a clean reference for downstream resistance interpretation; serial draws after first progression let the team distinguish on-pathway resistance (reversion) from off-pathway escape. Without a baseline the reversion call becomes ambiguous when the patient later progresses on olaparib or talazoparib.	Guardant Health (Guardant360 CDx) · test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887	5-10 mL plasma (Streck tube); Signatera also needs an FFPE block for assay design
ITWG stromal-TIL re-scoring on H and E by a dedicated breast pathologist (Salgado 2014 methodology)	The 3+ stromal-TIL label sits cleanly with the ITWG methodology only if it was scored as a continuous 0-100% percentage on a single H and E section by a pathologist trained on the Salgado 2014 guideline; ad-hoc 1+/2+/3+ grading is a known pitfall and inflates inter-observer variability. The re-score does not gate ICI eligibility but it is the reference standard the KEYNOTE-522 and ImPACT literature uses, and it makes the TIL signal defensible at tumor board.	Memorial Sloan Kettering Pathology Consultation Service · test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511	1 representative H and E slide; archival acceptable

MYC FISH (8q24 break-apart / amplification probe set) or MYC IHC (clone Y69)	MYC amplification by NGS copy-number is reasonably specific but the magnitude (low-level gain vs focal high-level amplification) varies by panel and tumor purity. FISH or IHC arbitrates this for the BET, CDK7, and CDK9 trial rationale, which generally favors high-level amplification or strong protein overexpression rather than borderline gain. The result refines trial eligibility for MYC-pathway protocols but does not gate any approved therapy.	NeoGenomics Laboratories (MYC FISH) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1 FFPE block or 4 unstained slides
B2M and HLA class I IHC plus B2M / HLA-A/B/C variant call review on tumor NGS	B2M loss and HLA class I antigen-presentation defects are well-characterized primary-resistance mechanisms for ICI in TMB-high and MSI-H tumors; loss of cell-surface MHC class I removes the substrate that PD-1 blockade depends on. In a TMB-14 / TIL-rich / MSS TNBC the ICI rationale is strong, but a B2M-deficient subclone would predict primary ICI failure even with the favorable TIL and TMB signals, and is detectable on the same FFPE block used for PD-L1 and TIL re-scoring. Cheap to add at the same time as the PD-L1 stain.	NeoGenomics Laboratories (B2M IHC / HLA Class I IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1-2 unstained slides; can share the PD-L1 / TIL block

Comprehensive NGS review for replication-stress / DDR co-alterations (ATM, ATR, CHK1/2, CDK12, CCNE1 amplification) and TP53 variant-functional classification	TP53 mutation is near-universal in basal-like TNBC and not directly drug-gating, but it anchors the replication-stress phenotype that WEE1 inhibitors (adavosertib, debio-0123) and ATR inhibitors (ceralasertib, elimusertib) exploit. The actionability hinges on co-alterations: CCNE1 amplification, CDK12 loss, or ATM loss meaningfully shift the WEE1 / ATR rationale, and TP53 missense vs nonsense vs hotspot R175H / R248Q / R273H functional class also affects experimental-trial enrollment.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with the PIK3CA / PTEN / TMB block
IRS2 RNA expression on tumor RNA-seq with reference to the TCGA TNBC distribution	IRS2 copy-number amplification only matters clinically if it translates to elevated IRS2 transcript and a functional IGF-1R / IRS signaling readout; many copy-number gains in TNBC do not. RNA-level confirmation is the cheapest way to triage whether IRS2 is worth chasing on an IGF-1R-axis trial versus parking it as a passenger event. Not gating any approved therapy; deferrable until an IGF-1R-axis protocol is on the table.	Tempus Labs (Tempus xR (whole-transcriptome RNA-seq)) · test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137	FFPE block or 10 unstained slides; can bundle with the PAM50 / Lehmann RNA-seq order

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.